



6-screen Video chat

Facebook Messenger is launching a new group chat feature where upto 6 different people can video chat at once. <http://dgit.in/FBnewVchat>



Sensitive Robot hands

At the Cornell University, researchers build a robotic hand that detect shape, softness and textures. <http://dgit.in/SenseBot>

Sensors of the Future

Gyroscopes and accelerometers are so last gen. Here's an in-depth look at the new wave of sensors that are likely to be at the heart of upcoming consumer devices. EEG/EOG sensors, arrays of swarm sensors, self-powered sensors and biodegradable sensors are just the tip of the iceberg - sensor technology is set to revolutionize industry, research and consumer electronics.

Aabir Abubaker
feedback@digit.in

The popular yardstick (though one that has lost relevance in the last decade) for progress in technology appears to have comfortably settled to Moore's Law. It's somewhat strange that this continues to be the popularly perceived rate of progress in the world of electronics, robotics and their applications. In fact, the obsession with processing power often shadows the equally exciting developments in algorithms, architecture, and the focus of this article - sensors.

What's a sensor good for?

Sensor technology had its humble origins in the tactile switch - yes, the very same button that you press to turn on the lights. The concept is a general one. A sensor must follow three very simple rules:

1. It must be sensitive to a property of interest
2. It must be insensitive to all other properties
3. It should not influence the property of interest

The logical progression of thought was - what if other inputs could be used to control similar outputs? The boom in aircraft technology called for a tremendous innovation in sensors in the 60s, with the fly-by-wire, hands-on control systems gradually being phased out in lieu of slicker digital interfaces. And the robotics boom (did you know that robots probably sort your mail?) put the cherry on top as we entered the 2000s. Even so, ask an average (tech-savvy!) kid on the street to name some sensor types and the answers you get will be limited to accelerometers, gyroscopes, and perhaps photodiodes (or 'light detectors', as goes teenage jargon).

What are we missing out on here? A whole lot, honestly. The 21st century has ushered in a new wave of sensor-based research that is set to completely change the way we think of technology and its capabilities. It's easy to see the problems with conventional sensors. They're bulky, heavy, fragile, need batteries, and are bad for the environment. Ignoring aesthetics, consider their functionality. Traditional sensors are limited to coarse measurements of

sound, light, speed and location - they lose out on chemical information like smell, on collaborative sensing, on very small inputs, and on most kinds of subtle information. Even worse, they often need to be treated with great care, and are next to useless in extreme conditions - imagine what you'd need to fly a drone over Mount Everest.

The 21st century, being the amazing period of time that it is, has seen fit to begin addressing all of these issues.

Micro- and Nano-sensors

Biological research has been the driving force behind a lot of modern sensor applications, and perhaps the best example is micro- and nano-sensor research. These are sensors with an extreme scaling down - literally of the order of micro or nanometres in length. To put things in perspective, the average cell is a few micrometres across. Even with current applications that are mainly tactile, micro-sensors and robots at such scales will give us the ability to treat diseases at the cellular level, to monitor wound healing to microcontrol drug dosage. In the future, they could